SANKSHEP.AI · SEVA FIRST INNOVATION CHALLENGE · TRACK 2 · TEAM TM003665

System architecture — *one source in, every format out*

A Gen-AI content transformation platform: the pipeline runs in the browser, and each model call goes to a cloud provider or to an on-device model, by the user's choice.

LEGEND
- **Private mode** · never leaves the device
- **Cloud mode** · AI requests
- **Accounts & data** · Supabase

USER'S DEVICE

BROWSER React 19 · TypeScript 5.7 · Vite 8 · Tailwind CSS 4 · installable PWA · 13-language interface · the whole pipeline runs here

WORKSPACE UI

UI Source & configuration
Upload PDF · DOCX · image, paste text or a link. Pick from 10 formats or describe one; tone and audience.

Processing mode: Cloud APIs | Private

Hardware check (RAM · cores · GPU) and Ollama setup guide before Private runs.

UI Output canvas
Collects every draft; on-device drafts stream in token by token.

Compare up to 3 models side by side · refine one draft in plain words · Stop at any moment.

Export PowerPoint (.pptx), Markdown or text.

UI History · Analytics · Admin
Every draft with its format, model and word counts; usage totals.

Admin: all users and activity; disable, enable or delete accounts.

Every `/workspace` route sits behind an auth guard.

CLIENT-SIDE INGESTION

Document readers
PDF → text (pdf.js), DOCX → text (mammoth), TXT · CSV · JSON · MD. Scans → page images.

Image reader
Vision model (Gemini, OpenRouter) or on-device OCR (Tesseract.js). Always OCR in Private mode.

Link reader
Page → text through Jina Reader. Cloud mode only.

AI ORCHESTRATION · LANGGRAPH-STYLE NODE GRAPH

N1 Ingest node — Merges text, file, link, image

N2 Parse node — Local clean-up, no AI call

N3 Fit source — Size budget per provider

N4 Format × Model — One node per (format, model), fanned out in parallel: 2–3 per provider, 1 on-device.
`tone · audience · instructions`

N5 Translate — Whole text, chunk by chunk · 13 languages

N6 Refine router — Draft only, or draft + source

N7 Run controller — Smart generation check skips drafts you already have.

One AbortController per run: Stop keeps finished drafts.

MODEL GATEWAY · CHAT()

G Route every call
Retries rate limits, 2-min timeout, own-key consent prompt before shared keys.

- **No own key** → /api/chat, shared keys
- **Own API key** → provider directly
- **Private mode** → Ollama on this computer

PRIVATE MODE **Local engine · Ollama · qwen2.5vl:7b** `localhost:11434`

Streamed /api/generate: drafts arrive token by token; Stop closes the stream at once.

num_ctx 4096 · temperature 0.3: fits an 8 GB laptop GPU, stays close to the source.

Zero data leaves: no server, no provider, no key. History keeps counts only.

CLOUD SERVICES

DB Supabase · accounts & data

Auth
Email sign-in and sessions, plus a demo account. Disabled accounts are banned here and signed out everywhere.

Postgres + RLS
`profiles` · `activity_logs`. Each user reads only their own rows; the admin reads all.

Server-checked functions: `admin_list_users` · `admin_set_user_disabled` · `admin_delete_user` · `my_account_disabled`

WEB Jina Reader
Fetches a linked page server-side and returns it as text, because models can't open links themselves.

API Vercel

Static hosting
The React app, PWA manifest and service worker.

`/api/chat` function
Holds the shared provider keys in environment variables; rotates them and retries rate limits.

AI Cloud AI providers — 11 drafting models

Groq	`gpt-oss-120b` (default) · `gpt-oss-20b` · `gpt-oss-safeguard-20b` · `qwen3.8-27b`. Fastest drafts.
Gemini	`3.5 Flash-Lite` · `3.8 Flash`. Reads text and images; the longest sources.
OpenRouter	Free models: `openrouter/free` · `Gemma 4 31B` · `Nemotron 3 Ultra` · `Nemotron 3.5 Lightning` · `Inkling`. Vision on the free router, Gemma and Inkling.

A small, fast model per provider routes each refinement. Users may paste their own key; it stays in the tab's memory and is sent only to that provider.

PRIVATE In Private mode none of this column sees the document: no text, images, drafts or file names reach the cloud.

10 formats	3 + 1 engines	4,096 tokens	13 languages	RLS on every table
+ any custom format, drafted in parallel	Groq · Gemini · OpenRouter, or Ollama on-device	local window, sized for an 8 GB GPU	English + 12 Indian, full-length translation	data access enforced by the database

How it works

The whole pipeline runs in the user's browser. One gateway then sends each AI request either to a cloud provider, for speed, or to a model on the same computer, when a document must never leave it.

1 The journey of one request

01
Add a source
Upload a PDF, Word file or image, paste text, or add a link.

→

02
Read it in the browser
Files become text on the user's device, before any AI call.

→

03
Draft every format
Each chosen format is written for the chosen tone and audience.

→

04
Review and refine
Compare models, ask for changes in plain words, then export.

→

05
Save to History
Each run is logged, so drafts can be found again later.

2 What each node does

labels match page 1

N1 Ingest
Gathers the pasted text, files, link and images into one source.

N2 Parse
Cleans the text on the device. No AI call, so nothing is cut off.

N3 Fit source
Trims long documents to what each model can read, sampling evenly.

N4 Format × Model
Writes one format with one model. Many run side by side.

N5 Translate
Translates the full document into English or any of 12 Indian languages, part by part.

N6 Refine router
Decides if an edit needs only the draft, or the source too.

N7 Run controller
Skips drafts that already exist, and stops everything on Stop.

G Gateway
Sends each AI request to the cloud, or to the on-device model.

3 Two ways to run

	Cloud APIs	Private, on-device
Who writes	Groq, Gemini, OpenRouter	qwen2.5vl:7b via Ollama
Your document	Sent to the provider	Never leaves the device
History keeps	An excerpt and drafts	Counts only
Best for	Speed and long sources	Confidential documents

4 Built-in safeguards

- **API keys stay on the server.** Shared keys never reach the browser.
- **Nothing switches silently.** If a user's key fails, we ask first.
- **Stop means stop.** One click cancels every request at once.
- **Each user sees only their own data.** The database enforces it.

5 Team TM003665

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Backend & data security

Built with React 19 · TypeScript · Vite · Tailwind CSS · Supabase · Vercel · Groq · Gemini · OpenRouter · Ollama